

VATTIKUTI SRI MAHESH ABHIRAM

AI Engineer — Research Intern

+91-630-996-5777 srimaheshabhiram@gmail.com India

linkedin.com/in/abhiramvattikuti github.com/Abhiram2005-pro

abhiram2005-pro.github.io/Portfolio

SUMMARY

Results-driven AI Engineer and Research Intern specializing in Deep Learning, Computer Vision, and NLP. Proven track record of architecting scalable Generative AI systems, multimodal pipelines, and LLM-powered applications. Passionate about deploying production-ready machine learning models to solve complex engineering challenges and drive technological innovation.

EXPERIENCE

AI Chatbot Developer

Zaytoon Solutions

06/2025 - 07/2025 Remote (Bangladesh)

- Architected an enterprise-grade NLP conversational banking chatbot, integrating backend SQL databases with LLMs to process over 1,000 daily financial inquiries
- Deployed a zero-shot localized translation pipeline leveraging DeepSeek, NLLB, and Bangla-LLaMA, reducing translation latency by 20% and improving multilingual accessibility
- Prototyped full-stack application wireframes in Gradio and migrated core components to a custom scalable frontend interface, boosting end-user engagement

Research Intern – Lip-to-Speech Synthesis

Amrita Vishwa Vidyapeetham

06/2025 - Present Coimbatore, India

- Designed and trained an end-to-end multimodal deep learning pipeline incorporating 3D CNNs and Transformer encoders, achieving 92% validation accuracy on the LRS2 dataset using CTC loss
- Constructed a Generative AI Visual Text-to-Speech (TTS) architecture enabling synchronized, high-fidelity speech audio synthesis
- Applied Real-Time Voice Cloning frameworks for personalized speaker embeddings, maximizing synthesized output speech quality across diverse profiles
- Curated and preprocessed a dedicated tracking dataset of 30 distinct speakers from 50+ hours of video/audio streams using OpenCV and automated data pipelines

EDUCATION

B.Tech in Artificial Intelligence and Data Science

Amrita Vishwa Vidyapeetham

08/2023 - 05/2027 Coimbatore, India

Intermediate Class XII Board Examination

Tirumala Junior College

06/2021 - 05/2023 Andhra Pradesh, India

Secondary School Certificate Class X

Tirumala Educational Institutes

06/2020 - 04/2021 Andhra Pradesh, India

KEY ACHIEVEMENTS

IEEE Publication – Co-Author

Co-authored "Identification and Sorting of Objects by Color using Robotic Arm" (2025 IEEE 6th Int. Conf. on Recent Advances in Information Technology - RAIT). Applied Computer Vision, CNN-based classification, Image Processing, and Robotics Automation for object identification and sorting.

LANGUAGES

English

Fluent

Telugu

Native

SKILLS

Python	C++	Java	C	SQL	PyTorch
TensorFlow	OpenCV	Transformers			
Deep Learning	Computer Vision	NLP	LLMs		
Generative AI	Agentic AI	Robotics			
Data Structures & Algorithms	Git				

PROJECTS

Selected AI & ML Projects

- 01/2023 - Present Remote
- Portfolio demonstrating applied R&D expertise in Artificial Intelligence and Software Engineering
- Developed an autonomous agricultural rover leveraging OpenCV and edge-deployed Computer Vision for real-time plant disease detection
 - Engineered AI-powered smart glasses featuring real-time object detection, DeepSort tracking, and voice-guided navigation
 - Fine-tuned CodeLLaMA foundational models to build an intelligent Agentic AI programming assistant for code generation
 - Customized Transformer architectures for English-to-Telugu neural machine translation (NMT), optimizing data preprocessing to improve BLEU scores
 - Built a Random Forest machine learning pipeline to identify catalytic interaction boundaries, accelerating data analysis
 - Designed a predictive network orchestration framework using supervised ML for intelligent resource allocation
 - Implemented an optimized decision tree search engine leveraging advanced Data Structures to achieve $O(\log n)$ retrieval efficiency